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CONCAVE FOR A TWIN ROTOR COMBINE HARVESTER

Abstract

A concave apparatus is provided for use in a twin rotor combine harvester which includes replaceable concave inserts which are carried in a concave support frame including two side rails which extend around a portion of each rotor from a bottom of the rotor outwardly around approximately 90 degrees. The frame includes transverse connecting bars at right angles to the side rails with an upper bar, a center bar and a lower bar defining between them openings for the concave inserts. An upper transverse bar of the lower concave insert has fastener holes to receive coupling fasteners for connection to the center transverse bar of the concave support frame and has a projecting portion lying in an axial plane and projecting radially outwardly to a position beyond the center transverse bar of the concave frame with fastener holes therethrough for attachment to pivotal bracket of the combine harvester.

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Background/Summary

BACKGROUND OF THE INVENTION

[0001] This invention relates to a concave apparatus for mounting on a combine harvester at one rotor of a twin rotor arrangement.

[0002] This invention relates to a combine harvester in which an axial flow rotary separator receives threshed material from a threshing cylinder and concave extending transversely with respect to the separator.

[0003] Combine harvesters using a rotary separator consisting of two side-by-side separator units receive threshed material from a cylinder and concave upstream of the separator are well known. This configuration is attractive and has become important in recently released new combine harvesters by important manufacturers including John Deere and CASE IH.

[0004] It combines the predictable and well understood crop performance of the conventional threshing cylinder and concave and the space-saving potential of the rotary separator.

[0005] Each of the two cylindrical rotors of the twin rotor arrangements includes a concave arrangement including a series of arcuate concave elements side by side, each extending around a portion of the cylinder which can be in an angle of the range 90 to 120 degrees. The side by side concave element are arranged at positions along the axis of the rotor so as to make up a portion of the length of the rotor.

[0006] The arrangement herein thus provides apparatus for mounting concaves in a combine having the pair of threshing and separating rotors extending in a longitudinal direction wherein the concaves can be mounted on its adjustable mounting frame through the side of the combine. A plurality of longitudinal spaced concave frame members are provided with yokes engageable with longitudinally extending mounting pins affixed to the mounting frame. Retaining links mounted on the mounting pins in conjunction with the concave yokes are connected to the concave frame members to retain the concave on the mounting pins during operation of the combine.

SUMMARY OF THE INVENTION

[0007] It is an object of the present invention to provide a concave arrangement for a dual rotor combine harvester which allows replacement of individual concave inserts while a concave frame remains in place. The arrangement herein is particularly designed and constructed for use with a John Deere X9 combine harvester. This arrangement allows the concaves to be changed for different crops or replaced when worn without the difficulty of removing the whole concave assembly.

[0008] According to a first aspect of the invention there is provided a concave apparatus for use in a combine harvester where the combine harvester includes: [0009] first and second side by side threshing and separating rotors; [0010] each rotor having associated therewith mounting arrangements for mounting a series of arcuate side by side concave elements, where each concave element extends angularly around a portion of the cylinder and where the side by side concave element are arranged at positions along the axis of the rotor; [0011] the mounting arrangement including an upper support for receiving an upper end of a respective concave element and a lower support for receiving a lower end of the respective concave element; [0012] the concave apparatus comprising: [0013] a concave support frame including two side rails arranged to lie in respective radial planes of the axis and extend around the portion of the cylinder and a plurality of transverse connecting bars at right angles to the side rails; [0014] a pair of concave inserts each for insertion into and mounting on the concave support frame; [0015] each concave insert having a rectangular arcuate body with a plurality of rub bars facing the rotor for cooperation therewith; [0016] the transverse connecting bars of the concave support frame including an upper bar, a center bar and a lower bar defining an upper rectangular space for an upper one of the pair of concave inserts and a lower rectangular space for a lower one of the pair of concave inserts; [0017] the upper concave insert having a lower transverse bar at a lower end and an upper transverse bar at an upper end; [0018] with the upper transverse bar of the upper concave insert having fastener holes to receive

coupling fasteners for connection to the upper transverse bar of the concave support frame; [0019] with the lower transverse bar of the upper concave insert having fastener holes to receive coupling fasteners for connection to the center transverse bar of the concave support frame; [0020] the lower concave insert having a lower transverse bar at a lower end and an upper transverse bar at an upper end; [0021] with the upper transverse bar of the lower concave insert having fastener holes to receive coupling fasteners for connection to the center transverse bar of the concave support frame; [0022] with the lower transverse bar of the lower concave insert having fastener holes to receive coupling fasteners for connection to the lower transverse bar of the concave support frame; [0023] at least one of the lower transverse bar of the upper concave insert and the upper transverse bar of the lower concave insert having a projecting portion lying in an axial plane of the axis and projecting in a direction radially outwardly from the axis to a position beyond the center transverse bar of the concave frame; [0024] the projecting portion having a plurality of fastener holes therethrough.

[0025] Preferably the upper transverse bar of the lower concave insert has the projecting portion thereon.

[0026] Preferably the upper and lower concave inserts are symmetrical for use in either the upper space or the lower space. Thus the upper insert also has the projecting portion which is not used in the mounting but allows the same component to be used in either location for simplicity or for replacement when worn.

[0027] Preferably each of the concave inserts includes side rails each of which lies immediately adjacent a respective side rail of the concave support rail.

[0028] Preferably the side rails of the concave insert are narrower than the concave support rails and the transverse bars of the concave support frame have a depth substantially equal to the side rails of the concave insert. Thus the projecting portion extends beyond the side rails of the concave insert.

[0029] Preferably the transverse connecting bars of the concave support frame including a top bar above the upper bar for attachment to the upper support of the mounting arrangement and wherein the concave support frame includes a lower engagement portion for engaging the lower support of the mounting arrangement.

[0030] Preferably the lower support of the mounting arrangement comprises a cylindrical rod extending parallel to the axis and the lower engagement portion comprises on each side rail a respective cup shaped receptacle for receiving the rod.

[0031] Preferably the upper support of the mounting arrangement comprises a V-shaped channel extending parallel to the axis and wherein the top bar of the concave support frame is arch shaped to engage into the V-shaped channel.

[0032] Preferably the concave support frame includes a concave portion between the upper bar and the top bar which has rub bars for cooperation with the rotor and the concave support frame includes a concave portion between the lower bar and the lower engagement portion which has rub bars for cooperation with the rotor.

[0033] According to a second aspect of the invention there is provided a combine harvester comprising: [0034] first and second side by side threshing and separating cylindrical rotors each rotatable about a respective axis of the rotor; [0035] a concave apparatus comprising a series of arcuate side by side concave elements [0036] each rotor having associated therewith mounting arrangements for mounting the side by side concave elements; [0037] where each concave element extends angularly around a portion of the cylindrical rotor; [0038] where the side by side concave element are arranged at positions along the axis of the rotor; [0039] the mounting arrangement including an upper support for receiving an upper end of a respective concave element and a lower support for receiving a lower end of the respective concave element; [0040] the mounting arrangement including for each respective concave element a bracket pivotally mounted at the lower support about an axis parallel to the axis and having a connecting bar movable between a

first position at the concave element and a second release position spaced from the concave element; [0041] each concave element comprising: [0042] a concave support frame including two side rails arranged to lie in respective radial planes of the axis and extend around the portion of the cylinder and a plurality of transverse connecting bars at right angles to the side rails; [0043] a pair of concave inserts each for insertion into and mounting on the concave support frame; [0044] each concave insert having a rectangular arcuate body with a plurality of rub bars facing the rotor for cooperation therewith; [0045] the transverse connecting bars of the concave support frame including an upper bar, a center bar and a lower bar defining an upper rectangular space for an upper one of the pair of concave inserts and a lower rectangular space for a lower one of the pair of concave inserts; [0046] the upper concave insert having a lower transverse bar at a lower end and an upper transverse bar at an upper end; [0047] with the upper transverse bar of the upper concave insert having fastener holes to receive coupling fasteners for connection to the upper transverse bar of the concave support frame; [0048] with the lower transverse bar of the upper concave insert having fastener holes to receive coupling fasteners for connection to the center transverse bar of the concave support frame; [0049] the lower concave insert having a lower transverse bar at a lower end and an upper transverse bar at an upper end; [0050] with the upper transverse bar of the lower concave insert having fastener holes to receive coupling fasteners for connection to the center transverse bar of the concave support frame; [0051] with the lower transverse bar of the lower concave insert having fastener holes to receive coupling fasteners for connection to the lower transverse bar of the concave support frame; [0052] at least one of the lower transverse bar of the upper concave insert and the upper transverse bar of the lower concave insert having a projecting portion lying in an axial plane of the axis and projecting in a direction radially outwardly from the axis to a position beyond the center transverse bar of the concave frame; [0053] the projecting portion having a plurality of fastener holes therethrough for fastening to the connecting bar when in the first position at the concave element; [0054] the second release portion of the bracket being arranged such that the connecting bar is spaced from the concave element to allow release of the concave insert from the concave support frame while the concave support frame remains in place attached to the upper and lower supports.

[0055] Preferably the bracket includes two side rails connected to and supporting the connecting bar where the side rails of the bracket lie immediately adjacent the side rails of the concave support frame.

[0056] Preferably the lower support of the mounting arrangement comprises a cylindrical rod extending parallel to the axis and the side rails of the bracket pivot around the cylindrical rod.

[0057] According to a third aspect of the invention there is provided a combine harvester comprising: [0058] first and second side by side threshing and separating cylindrical rotors each rotatable about a respective axis of the rotor; [0059] a concave apparatus comprising a series of arcuate side by side concave elements [0060] each rotor having associated therewith mounting arrangements for mounting the side by side concave elements; [0061] where each concave element extends angularly around a portion of the cylindrical rotor; [0062] where the side by side concave element are arranged at positions along the axis of the rotor; [0063] the mounting arrangement including an upper support for receiving an upper end of a respective concave element and a lower support for receiving a lower end of the respective concave element; [0064] the mounting arrangement including for each respective concave element a bracket pivotally mounted at the lower support about an axis parallel to the axis and having a bracket connecting bar movable between a first position at the concave element and a second release position spaced from the concave element; [0065] each concave element comprising: [0066] a concave support frame including two side rails arranged to lie in respective radial planes of the axis and extend around the portion of the cylinder and a plurality of transverse connecting bars at right angles to the side rails; [0067] a pair of concave inserts each for insertion into and mounting on the concave support frame; [0068] each concave insert having a rectangular arcuate body with a plurality of rub bars facing the

rotor for cooperation therewith; [0069] the transverse connecting bars of the concave support frame including an upper bar, a center bar and a lower bar defining an upper rectangular space for an upper one of the pair of concave inserts and a lower rectangular space for a lower one of the pair of concave inserts; [0070] the upper concave insert having a lower transverse bar at a lower end and an upper transverse bar at an upper end; [0071] with the upper transverse bar of the upper concave insert having fastener holes to receive coupling fasteners for connection to the upper transverse bar of the concave support frame; [0072] with the lower transverse bar of the upper concave insert having fastener holes to receive coupling fasteners for connection to the center transverse bar of the concave support frame; [0073] the lower concave insert having a lower transverse bar at a lower end and an upper transverse bar at an upper end; [0074] with the upper transverse bar of the lower concave insert having fastener holes to receive coupling fasteners for connection to the center transverse bar of the concave support frame; [0075] with the lower transverse bar of the lower concave insert having fastener holes to receive coupling fasteners for connection to the lower transverse bar of the concave support frame; [0076] the bracket connecting bar being connected to the respective concave element when in the first position at the concave element; [0077] the bracket connecting bar being releasable from the respective concave element to move to the second release position of the bracket such that the bracket connecting bar is spaced from the concave element to allow release of the concave insert from the concave support frame while the concave support frame remains in place attached to the upper and lower supports.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0078] One embodiment of the invention will now be described in conjunction with the accompanying drawings in which:

[0079] FIG. 1 is a left side elevational view of a PRIOR ART combine harvester to illustrate the components of the combine harvester with which the present invention cooperates, a portion of the side sheet of the combine being broken away to more clearly show the relative positions of the threshing and separating rotors and cooperatively mounted concave.

[0080] FIG. 2 is an enlarged partial cross-sectional view taken along lines 2-2 of the PRIOR ART combine harvester of FIG. 1, looking along the longitudinally extending axis of the threshing and separating rotor, the portions of the infeed structure in front of the left hand rotor being broken away to more clearly show the structure and relationships utilized in the side mount concaves forming the instant invention.

[0081] FIG. 3 is an isometric view of the inside face and one side of the concave apparatus according to the invention for mounting in the combine harvester of FIGS. 1 and 2.

[0082] FIG. 4 is an isometric view of the outside face and one side of the concave apparatus of FIG. 3 according to the invention for mounting in the combine harvester of FIGS. 1 and 2.

[0083] FIG. 5 is an isometric view of the outside face and one side of the concave apparatus according to the invention including components of the combine harvester with which it cooperates where the arrangement is shown in the release position in which the concave inserts can be removed from the frame.

[0084] FIG. 6 is a cross-sectional view of the concave apparatus according to the invention including components of the combine harvester with which it cooperates showing the arrangement in the release position.

[0085] FIG. 7 is an isometric view of the outside face and one side of the concave apparatus according to the invention including components of the combine harvester with which it cooperates where the arrangement is shown in the locked position in which the concave inserts are fixed in the frame for operation.

[0086] FIG. **8** is a cross-sectional view of the concave apparatus according to the invention including components of the combine harvester with which it cooperates where the arrangement is shown in the locked position in which the concave inserts are fixed in the frame for operation.

[0087] FIG. **9** is an exploded view of the components shown in the position of FIG. **3**.

[0088] FIG. **10** is a cross-sectional view on an enlarged scale showing the cooperation between the bracket and the concave inserts.

DETAILED DESCRIPTION

[0089] Referring now to the drawings and, particularly, to FIG. **1**, a left side elevational view of a crop harvesting machine can be seen. The combine **10** is mounted on a wheeled frame **12** to move the combine **10** over the ground **G** for the harvesting of crops therefrom. A header **13** is mounted on the forward periphery of the combine **10** to engage and gather crop material from the ground and consolidate it into a feederhouse **14**. The feederhouse **14** conveys a flow of crop material into the infeed area **16** of the threshing and separating rotor **15**. The flow of crop material then passes into the threshing area **17** between the rotor **15** and the circumferentially mounted concave **20** where grain is threshed and separated from the crop material by rasp bar assemblies **18** cooperating with the concave **20**.

[0090] Referring now to FIG. **2**, the relationship of the rotor **15** and the rasp bar assemblies **18** mounted thereon with respect to the concave **20** can best be seen. The concave **20** is provided with at least two longitudinally spaced side rails **22**, the forwardmost of which is designated by the reference numeral **23**, which carry a plurality of longitudinally extending rub bars **24** radially positioned relative to the rotor **15**. Crop material entering between the rotor **15** and the concave **20** is rubbed between the rasp bar assemblies **18** and the rub bars **24** to separate the grain from the crop material. The concave **20** is mounted on an adjustable mounting frame **25** to permit movement of the concave **20** relative to the rotor **15**.

[0091] In the remaining FIGS. **3** to **10** is shown a concave apparatus for use in a combine harvester of the type shown in FIGS. **1** and **2**.

[0092] In this arrangement, the concave apparatus is arranged to extend from a position adjacent the bottom of the rotor over an angle of around 90 to 120 degrees to a position outwardly of the rotor. Thus each concave apparatus is only accessible from the respective outer side of the each of the rotors.

[0093] The concave apparatus includes a plurality of the components shown in the figures which are arranged side by side along the rotor to cover a portion only of the length of the rotor as shown in FIG. **1**. The individual concave components can thus be removed from the mounting arrangement which holds them in place in the combine as described hereinafter. The number and width of the concave components can vary to make up together the required length. The components can be removed and replaced to change the rub bar effect for different crops and for replacement when worn.

[0094] As shown in FIGS. **3** and **9**, the concave apparatus **30** includes a concave support frame **31** including two arcuate side rails **32** and **33** arranged to lie in respective radial planes of the axis of the rotor and extend around the portion of the rotor as required. The frame **31** further includes a plurality of transverse connecting bars at right angles to the side rails **32** and **33**. These include an upper bar **34**, a center bar **35** and a lower bar **36** spaced to define an upper rectangular space **37** for an upper one of a pair of concave inserts **38** and a lower rectangular space for a lower one of the pair of concave inserts;

[0095] The concave inserts shown at **39** and **40** comprise a pair of the inserts each for insertion into and mounting on the concave support frame **31** at the spaces **37** and **38**.

[0096] The pair of inserts have the same construction so that they are symmetrical and thus each can be mounted in each of the spaces. In this way they can be supplied as a common part reducing inventory and also can be replaced one for the other if wear occurs. Each concave insert **39**, **40** has a rectangular arcuate body with a pair of side rails **42** and plurality of rub transverse bars **41** facing

the rotor for cooperation therewith. Each concave insert has a width matching that of the space defined by the rails **32** and **33** so that the side rails **42** slide just inside the side rails **42** to fit therein without movement. The concave inserts each include a lower transverse bar **43** at a lower end and an upper transverse bar **44** at an upper end with the space between the bars **43** and **44** being arranged to fit in the space between the upper bar **34** and the center bar **35** or the space between the center bar **35** and the lower bar **36** of the concave frame.

[0097] The upper transverse bar **44** of the concave inserts has fastener holes **441** and **442** to receive coupling fasteners **45** for connection to the upper transverse bar **34** of the concave support frame **31**. The lower transverse bar of the concave inserts has fastener holes **451** and **452** to receive coupling fasteners **46** for connection to the center transverse bar **35** of the concave support frame **31**. Symmetrically, the lower concave insert has in the upper transverse bar fastener holes **441** and **442** to receive coupling fasteners for connection to the center transverse bar **35** of the concave support frame and in the lower transverse bar of the lower concave insert fastener holes **451** and **452** to receive coupling fasteners **47** for connection to the lower transverse bar **36** of the concave support frame **31**.

[0098] It will be noted that the holes **441** and **442** are smaller than the holes **451** and **452**. This is because the holes **441** and **442** directly receive a threaded fastener **452** of the fastener **45** as a sliding fit therein with nuts **453** connected on the upper face of the upper bar **34** of the frame to hold the upper end of the insert in place. Symmetrically the lower insert has the smaller holes in the lower bar so as to receive the threaded section **461** of the fastener **46**. The holes **451** and **452** are larger because they fit over the nut **471** and over a collar **472** of the fastener **47**. In this way each of the inserts is simply a sliding fit over the nut at its bottom bar and is screw fastened at the top bar. In this way the inserts are held in place by screw connectors only at the top and can be loosely coupled at the bottom so that when the upper screw fasteners are removed the insert can simply be pulled out from its place by pulling the bottom holes **451** and **452** off the collar **472**. A center hole **443** is provided between the holes **441** and **442** to receive an alignment punch prior to insertion of the fasteners **45** and **46** through the aligned holes in the bars to be coupled.

[0099] The upper transverse bar **44** of the lower concave insert has a projecting portion **444** lying in an axial plane of the axis of the rotor and projecting in a direction radially outwardly from the axis to a position beyond the center transverse bar **35** of the concave frame **31**. This is best shown in FIG. **10** but also can be seen in respect of the upper concave insert **39** in FIG. **9**, bearing in mind that the inserts are identical.

[0100] The projecting portion **444** having a plurality of fastener holes **445** therethrough at spaced positions across the projecting portion.

[0101] Turning now to FIGS. **5** and **6**, the mounting arrangement by which the concave is mounted in the combine harvester is shown. Thus the combine harvester includes an upper support beam **50** for receiving an upper end of a respective concave element and a lower support **51** for receiving a lower end of the respective concave element.

[0102] The beam **50** includes a V-shaped receptacle in its underside for receiving a top surface of a top plate **301** which fits into the receptacle and is fastened thereto by fasteners **502**. The beam **50** is a permanent mounting point on the combine harvester and is located at a suitable position along the rotor to carry the required number of side by side concave elements.

[0103] The lower support **51** comprises a rod extending along the rotor at an angularly spaced position relative to the beam **50** to define the space for the concave elements. The rod also is fixed permanently in position as part of the combine harvester. The lower end of the side rails **32** and **33** of the concave frame **31** has a part cylindrical receptacle **511** which engages around and receives the rod to hold the frame in place at the lower end. The side rails **32** and **33** are braced at the lower end by a cross bar **512**.

[0104] The arrangement of the combine harvester concerned also includes for each respective concave element a bracket **52** pivotally mounted at the lower support rod **51** about an axis along the

rod and thus parallel to the rotor axis. The bracket **52** includes two side rails **521** and **522** with an inner end pivotally mounted on the rod and an outer end carrying a connecting bar **523** extending across between the two side rails. The bracket **52** is movable between a first position, at the concave element as shown in FIGS. **7** and **8** where the bracket acts to hold the concave in place, and a second release position, shown in FIGS. **5**, **6** and **10** where the bracket and particularly the bar **523** is spaced from the concave element.

[0105] The second release position of the bracket is thus arranged such that the connecting bar **523** is spaced from the concave element to allow release of the lower concave insert **40** from the concave support frame **31** while the concave support frame **31** remains in place attached to the upper support beam **50** and the lower support rod **51**.

[0106] As best shown in FIG. **10**, when in the first position, the bar **523** is moved to a position along arrow **524** so that the bar is aligned with and parallel to the projecting portion **444**. In this position the bar **523** is bolted to the projecting portion through holes **445** using fasteners **525**.

[0107] As the projecting portion **444** extends radially outwardly beyond the center plate **35**, the bar **523** is bolted only to the projecting portion of the insert and not to the frame. This allows the bracket to be moved to the release position and the replaceable insert removed for replacement while the frame remains in place.

[0108] Each of the concave inserts includes the side rails **42** each of which lies immediately adjacent a respective side rail **32**, **33** of the concave support. The side rails **42** of the concave insert are narrower in a radial direction than the concave support rails **32**, **33** and the transverse bars **34**, **35** and **36** of the concave support frame have a depth substantially equal to the side rails **42** of the concave insert. In this way the projecting portion **444** extends beyond the side rails **42** of the concave insert.

[0109] The concave support frame as best shown in FIG. **9** includes a concave portion **307** between the upper bar **34** and the upper support beam **301** which has rub bars **309** for cooperation with the rotor. Also a concave portion **308** is provided between the lower bar **36** and the brace cross bar **512** which has rub bars **309**.

[0110] Since various modifications can be made in my invention as herein above described, and many apparently widely different embodiments of same made within the spirit and scope of the claims without departure from such spirit and scope, it is intended that all matter contained in the accompanying specification shall be interpreted as illustrative only and not in a limiting sense.

Claims

1. A concave apparatus for use in a combine harvester where the combine harvester includes: first and second side by side threshing and separating rotors; each rotor having associated therewith mounting arrangements for mounting a series of arcuate side by side concave elements, where each concave element extends angularly around a portion of the cylinder and where the side by side concave element are arranged at positions along the axis of the rotor; the mounting arrangement including an upper support for receiving an upper end of a respective concave element and a lower support for receiving a lower end of the respective concave element; the concave apparatus comprising: a concave support frame including two side rails arranged to lie in respective radial planes of the axis and extend around the portion of the cylinder and a plurality of transverse connecting bars at right angles to the side rails; a pair of concave inserts each for insertion into and mounting on the concave support frame; each concave insert having a rectangular arcuate body with a plurality of rub bars facing the rotor for cooperation therewith; the transverse connecting bars of the concave support frame including an upper bar, a center bar and a lower bar defining an upper rectangular space for an upper one of the pair of concave inserts and a lower rectangular space for a lower one of the pair of concave inserts; the upper concave insert having a lower transverse bar at a lower end and an upper transverse bar at an upper end; with the upper transverse

bar of the upper concave insert having fastener holes to receive coupling fasteners for connection to the upper transverse bar of the concave support frame; with the lower transverse bar of the upper concave insert having fastener holes to receive coupling fasteners for connection to the center transverse bar of the concave support frame; the lower concave insert having a lower transverse bar at a lower end and an upper transverse bar at an upper end; with the upper transverse bar of the lower concave insert having fastener holes to receive coupling fasteners for connection to the center transverse bar of the concave support frame; with the lower transverse bar of the lower concave insert having fastener holes to receive coupling fasteners for connection to the lower transverse bar of the concave support frame; at least one of the lower transverse bar of the upper concave insert and the upper transverse bar of the lower concave insert having a projecting portion lying in an axial plane of the axis and projecting in a direction radially outwardly from the axis to a position beyond the center transverse bar of the the projecting portion having a plurality of fastener holes therethrough.

2. The concave apparatus according to claim 1 wherein the upper transverse bar of the lower concave insert has the projecting portion thereon.

3. The concave apparatus according to claim 1 wherein upper and lower concave inserts are symmetrical for use in either the upper space or the lower space.

4. The concave apparatus according to claim 1 wherein each of the concave inserts includes side rails each of which lies immediately adjacent a respective side rail of the concave support rail.

5. The concave apparatus according to claim 4 wherein the side rails of the concave insert are narrower than the concave support rails.

6. The concave apparatus according to claim 4 wherein the transverse bars of the concave support frame have a depth substantially equal to the side rails of the concave insert.

7. The concave apparatus according to claim 4 wherein the projecting portion extends beyond the side rails of the concave insert.

8. The concave apparatus according to claim 1 wherein the transverse connecting bars of the concave support frame including a top bar above the upper bar for attachment to the upper support of the mounting arrangement and wherein the concave support frame includes a lower engagement portion for engaging the lower support of the mounting arrangement.

9. The concave apparatus according to claim 8 wherein the lower support of the mounting arrangement comprises a cylindrical rod extending parallel to the axis and the lower engagement portion comprises on each side rail a respective cup shaped receptacle for receiving the rod.

10. The concave apparatus according to claim 1 wherein the upper support of the mounting arrangement comprises a V-shaped channel extending parallel to the axis and wherein the top bar of the concave support frame is arch shaped to engage into the V-shaped channel.

11. The concave apparatus according to claim 1 wherein the concave support frame includes a concave portion between the upper bar and the upper support which has rub bars for cooperation with the rotor.

12. The concave apparatus according to claim 1 wherein the concave support frame includes a concave portion between the lower bar and the lower support which has rub bars for cooperation with the rotor.

13. A combine harvester comprising: first and second side by side threshing and separating cylindrical rotors each rotatable about a respective axis of the rotor; a concave apparatus comprising a series of arcuate side by side concave elements each rotor having associated therewith mounting arrangements for mounting the side by side concave elements; where each concave element extends angularly around a portion of the cylindrical rotor; where the side by side concave element are arranged at positions along the axis of the rotor; the mounting arrangement including an upper support for receiving an upper end of a respective concave element and a lower support for receiving a lower end of the respective concave element; the mounting arrangement including for each respective concave element a bracket pivotally mounted at the lower support about an axis

parallel to the axis and having a bracket connecting bar movable between a first position at the concave element and a second release position spaced from the concave element; each concave element comprising: a concave support frame including two side rails arranged to lie in respective radial planes of the axis and extend around the portion of the cylinder and a plurality of transverse connecting bars at right angles to the side rails; a pair of concave inserts each for insertion into and mounting on the concave support frame; each concave insert having a rectangular arcuate body with a plurality of rub bars facing the rotor for cooperation therewith; the transverse connecting bars of the concave support frame including an upper bar, a center bar and a lower bar defining an upper rectangular space for an upper one of the pair of concave inserts and a lower rectangular space for a lower one of the pair of concave inserts; the upper concave insert having a lower transverse bar at a lower end and an upper transverse bar at an upper end; with the upper transverse bar of the upper concave insert having fastener holes to receive coupling fasteners for connection to the upper transverse bar of the concave support frame; with the lower transverse bar of the upper concave insert having fastener holes to receive coupling fasteners for connection to the center transverse bar of the concave support frame; the lower concave insert having a lower transverse bar at a lower end and an upper transverse bar at an upper end; with the upper transverse bar of the lower concave insert having fastener holes to receive coupling fasteners for connection to the center transverse bar of the concave support frame; with the lower transverse bar of the lower concave insert having fastener holes to receive coupling fasteners for connection to the lower transverse bar of the concave support frame; at least one of the lower transverse bar of the upper concave insert and the upper transverse bar of the lower concave insert having a projecting portion lying in an axial plane of the axis and projecting in a direction radially outwardly from the axis to a position beyond the center transverse bar of the concave frame; the projecting portion having a plurality of fastener holes therethrough for fastening to the bracket connecting bar when in the first position at the concave element; the second release position of the bracket being arranged such that the bracket connecting bar is spaced from the concave element to allow release of the concave insert from the concave support frame while the concave support frame remains in place attached to the upper and lower supports.

14. The combine harvester according to claim 13 wherein the bracket includes two side rails connected to and supporting the bracket connecting bar where the side rails of the bracket lie immediately adjacent the side rails of the concave support frame.

15. The combine harvester according to claim 14 wherein the lower support of the mounting arrangement comprises a cylindrical rod extending parallel to the axis and the side rails of the bracket pivot around the cylindrical rod.

16. The combine harvester according to claim 13 wherein the upper transverse bar of the lower concave insert has the projecting portion thereon.

17. The combine harvester according to claim 13 wherein upper and lower concave inserts are symmetrical for use in either the upper space or the lower space.

18. The combine harvester according to claim 13 wherein each of the concave inserts includes side rails each of which lies immediately adjacent a respective side rail of the concave support rail; wherein the transverse bars of the concave support frame have a depth substantially equal to the side rails of the concave insert and wherein the projecting portion extends beyond the side rails of the concave insert.

19. The combine harvester according to claim 13 wherein the transverse connecting bars of the concave support frame including a top bar above the upper bar for attachment to the upper support of the mounting arrangement wherein the upper support of the mounting arrangement comprises a V-shaped channel extending parallel to the axis and wherein the top bar of the concave support frame is arch shaped to engage into the V-shaped channel.

20. A combine harvester comprising: first and second side by side threshing and separating cylindrical rotors each rotatable about a respective axis of the rotor; a concave apparatus

comprising a series of arcuate side by side concave elements each rotor having associated therewith mounting arrangements for mounting the side by side concave elements; where each concave element extends angularly around a portion of the cylindrical rotor; where the side by side concave element are arranged at positions along the axis of the rotor; the mounting arrangement including an upper support for receiving an upper end of a respective concave element and a lower support for receiving a lower end of the respective concave element; the mounting arrangement including for each respective concave element a bracket pivotally mounted at the lower support about an axis parallel to the axis and having a bracket connecting bar movable between a first position at the concave element and a second release position spaced from the concave element; each concave element comprising: a concave support frame including two side rails arranged to lie in respective radial planes of the axis and extend around the portion of the cylinder and a plurality of transverse connecting bars at right angles to the side rails; a pair of concave inserts each for insertion into and mounting on the concave support frame; each concave insert having a rectangular arcuate body with a plurality of rub bars facing the rotor for cooperation therewith; the transverse connecting bars of the concave support frame including an upper bar, a center bar and a lower bar defining an upper rectangular space for an upper one of the pair of concave inserts and a lower rectangular space for a lower one of the pair of concave inserts; the upper concave insert having a lower transverse bar at a lower end and an upper transverse bar at an upper end; with the upper transverse bar of the upper concave insert having fastener holes to receive coupling fasteners for connection to the upper transverse bar of the concave support frame; with the lower transverse bar of the upper concave insert having fastener holes to receive coupling fasteners for connection to the center transverse bar of the concave support frame; the lower concave insert having a lower transverse bar at a lower end and an upper transverse bar at an upper end; with the upper transverse bar of the lower concave insert having fastener holes to receive coupling fasteners for connection to the center transverse bar of the concave support frame; with the lower transverse bar of the lower concave insert having fastener holes to receive coupling fasteners for connection to the lower transverse bar of the concave support frame; the bracket connecting bar being connected to the respective concave element when in the first position at the concave element; the bracket connecting bar being releasable from the respective concave element to move to the second release position of the bracket such that the bracket connecting bar is spaced from the concave element to allow release of the concave insert from the concave support frame while the concave support frame remains in place attached to the upper and lower supports.

21. The concave apparatus according to claim 1 wherein the fastener holes of the lower transverse bar are arranged as a sliding fit over a collar of the coupling fastener so that each of the upper and lower concave inserts is simply a sliding fit over the collar at the lower end and is screw fastened at the upper end so that the upper and lower concave inserts are held in place by screw connectors only at the upper end and are loosely coupled at the lower end so that when the upper screw connectors are removed the concave insert can simply be pulled out from its place.

22. The combine harvester according to claim 13 wherein the fastener holes of the lower transverse bar are arranged as a sliding fit over a collar of the coupling fastener so that each of the upper and lower concave inserts is simply a sliding fit over the collar at the lower end and is screw fastened at the upper end so that the upper and lower concave inserts are held in place by screw connectors only at the upper end and are loosely coupled at the lower end so that when the upper screw connectors are removed the concave insert can simply be pulled out from its place.

23. A combine harvester comprising: first and second side by side threshing and separating cylindrical rotors each rotatable about a respective axis of the rotor; a concave apparatus comprising a series of arcuate side by side concave elements each rotor having associated therewith mounting arrangements for mounting the side by side concave elements; where each concave element extends angularly around a portion of the cylindrical rotor; where the side by side concave element are arranged at positions along the axis of the rotor; the mounting arrangement including

an upper support for receiving an upper end of a respective concave element and a lower support for receiving a lower end of the respective concave element; each concave element comprising: a concave support frame including two side rails arranged to lie in respective radial planes of the axis and extend around the portion of the cylinder and a plurality of transverse connecting bars at right angles to the side rails; a pair of concave inserts each for insertion into and mounting on the concave support frame; each concave insert having a rectangular arcuate body with a plurality of rub bars facing the rotor for cooperation therewith; the transverse connecting bars of the concave support frame including an upper bar, a center bar and a lower bar defining an upper rectangular space for an upper one of the pair of concave inserts and a lower rectangular space for a lower one of the pair of concave inserts; the upper concave insert having a lower transverse bar at a lower end and an upper transverse bar at an upper end; with the upper transverse bar of the upper concave insert having fastener holes to receive coupling fasteners for connection to the upper transverse bar of the concave support frame; with the lower transverse bar of the upper concave insert having fastener holes to receive coupling fasteners for connection to the center transverse bar of the concave support frame; the lower concave insert having a lower transverse bar at a lower end and an upper transverse bar at an upper end; with the upper transverse bar of the lower concave insert having fastener holes to receive coupling fasteners for connection to the center transverse bar of the concave support frame; with the lower transverse bar of the lower concave insert having fastener holes to receive coupling fasteners for connection to the lower transverse bar of the concave support frame; wherein the fastener holes of the lower transverse bar are arranged as a sliding fit over a collar of the coupling fastener so that each of the upper and lower concave inserts is simply a sliding fit over the collar at the lower end and is screw fastened at the upper end so that the upper and lower concave inserts are held in place by screw connectors only at the upper end and are loosely coupled at the lower end so that when the upper screw connectors are removed the concave insert can simply be pulled out from its place.
